

# MAHMOUD TAREK ELKADY

## Senior XR Engineer & Gameplay Systems Developer

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### PROFESSIONAL SUMMARY

Senior XR Engineer and Gameplay Systems Developer with over 3 years of commercial experience architecting VR/AR training simulations, modular gameplay systems, and multiplayer mechanics in Unity3D and Unreal Engine. Proven track record of delivering enterprise-grade clinical and industrial simulations (high-voltage electrical safety, medical response) using custom VR interaction toolkits and allocation-free C# architectures. IEEE-published researcher in AI/AR systems with a competitive esports background that drives a relentless focus on game feel, frame-rate optimization, and precise tick timing.

### TECHNICAL SKILLS

**Engines & XR Toolkits:** Unity3D, Unreal Engine (C++/Blueprints), OpenXR, XR Interaction Toolkit (XRI), Auto Hand, AR Foundation, ARCore, ARKit, WebGL.

**Programming & Architecture:** C#, C++, Python, Java, JavaScript, Object-Oriented Programming (OOP), Entity Component System (ECS), MVC, Data Structures & Algorithms.

**Unity Frameworks & Libraries:** UniTask (Async/Await), Zenject / Extenject (Dependency Injection), Addressables (Remote Content), TextMeshPro, Mirror Networking, NUnit (Unit / Play Mode Tests).

**DevOps & Tools:** Git, GitHub Actions, Jira, Confluence, Notion, OpenGL, Shader Graph / SDF.

### WORK EXPERIENCE

**Part-time Senior XR Engineer | Bluebell** January 2026 – Present

- Architected scalable VR systems and reusable interaction frameworks in Unity3D for high-performance enterprise training simulations.
- Engineered modular physics-based interaction mechanics, streamlining cross-project simulation pipelines and reducing new scenario development time.
- Partnered with cross-functional technical leads to design robust, decoupled architectures capable of handling complex simulation states.

**XR Developer | Merse** January 2024 – Present

- Developed immersive VR/AR training experiences utilizing Unity3D, XR Interaction Toolkit, and Auto Hand for enterprise client projects.
- Designed and built a lightweight, custom asynchronous timing system replacing Unity coroutines to eliminate garbage collection allocations and stabilize target frame rates on standalone VR headsets.
- Executed performance profiling, user testing, and frame-budget optimizations to ensure low-latency hand-tracking and motion interaction.

**Game Developer | Ekson Almotahida Education Group** April 2023 – January 2024

- Designed and built interactive 3D scientific virtual labs and educational mini-games from concept to deployment using Unity and Unreal Engine.
- Implemented core game loops, physics simulations, and responsive user interfaces to enhance learning retention in STEM subjects.

**Game Development Instructor | SIMPLIFE & GOMYCODE** April 2023 – January 2024

- Authored 2D/3D game development curricula covering C# scripting, game mechanics, and AR/VR integration.
- Mentored over 50+ students through hands-on game design projects, code reviews, and Git version control workflows.

## FEATURED PROJECTS

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### TVTC VR Electrical Safety Simulator | *Unity, OpenXR, C#*

- Developed a bilingual VR training simulation enabling technicians to practice high-voltage electrical safety and complex panel wiring procedures.

### Clinical Simulator Training XR & Medical Scenarios | *Unity, Hand-Tracking, VR*

- Built VR medical training applications (Clinical Training, Asthma Attack First Aid, Altitude Sickness) featuring realistic hand-tracking mechanics and symptom simulation.

### Color Swap Mobile Game | *Unity, C#, Mobile UI*

- Programmed a portrait-oriented mobile puzzle game featuring graph-coloring constraints, dynamic board generation, and adaptive UI layouts.

### Augmented Reality Multiplayer RPG (MAHO) | *Graduation Project — Unity, AR Foundation, Mirror*

- Integrated Google ARCore/ARKit with Unity AR Foundation and built real-time network synchronization via Mirror Networking API. Implemented Deep Reinforcement Learning for adaptive NPC behavior.

### 2D MMORPG & Survival Action Games | *Unity C#, Mirror Networking*

- Developed networked combat, custom NPC AI spawning algorithms, and inventory/farming systems with state-machine architectures.

## PUBLICATIONS & RESEARCH

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### Deep Reinforcement Learning in Augmented Reality Games | *IEEE Published* Dec 2020 – May 2021

- Conducted research on applying Deep Reinforcement Learning (DRL) techniques to AR environments, creating adaptive NPC behaviors to increase player engagement.

## EDUCATION

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### Postgraduate Diploma in Game Development | *Information Technology Institute (ITI)* Oct 2021 – Jan 2022

### Bachelor's Degree in Computer Science | *Misr International University (MIU)* 2017 – 2021

## COMPETITIVE ESPORTS & ACTIVITIES

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- **Professional Esports Athlete:** Competed at top regional tiers in *League of Legends* (Team Wild Panthers in Greek League, Team Anubis in Egyptian League). Applied competitive understanding of frame data, tick rates, and animation canceling directly into gameplay system design.
- **E-Sports Coach:** Sky Coach (Oct 2019 – Mar 2020).
- **Gamers Legacy Club Member:** Organized competitive gaming tournaments and game development panel showcases at MIU and Egycon (2018 – 2021).